

CS 4/5782 DoRA Project

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Github: <https://github.com/ishitasinha8/DoRAImplementation>

Introduction:

Large language models often require expensive full fine-tuning to adapt well to downstream tasks, making parameter-efficient fine-tuning methods important for resource-limited settings. Our project reproduces “DoRA: Weight-Decomposed Low-Rank Adaptation” by Shih-Yang Liu et al., which proposes improving LoRA by separating each pretrained weight into magnitude and direction components. DoRA applies low-rank adaptation to the weight direction while learning a separate magnitude vector, aiming to close the performance gap between LoRA and full fine-tuning while keeping the trainable parameter count close to LoRA and introducing no additional inference overhead after merging.

In this project, we implemented DoRA from scratch for a Llama-style language model and evaluated it on commonsense reasoning benchmarks. Our focus was not only on matching the paper’s reported results, but also on testing choices not emphasized in the paper, such as adapting only attention layers versus adding MLP projections, varying DoRA rank and sequence length, and comparing free-form generation evaluation against likelihood-based multiple-choice scoring.

Chosen Result

We aimed to reproduce the paper’s central commonsense reasoning result that DoRA improves over standard LoRA while retaining similar parameter efficiency. It reports accuracy across eight commonsense reasoning datasets: BoolQ, PIQA, SIQA, HellaSwag, WinoGrande, ARC-e, ARC-c, and OpenBookQA. This result was chosen because it directly evaluates the paper’s main claim that separating weight magnitude and direction improves adaptation quality.

In our implementation, we reproduced the DoRA adaptation pipeline from scratch and evaluated it on both base Llama and Llama-Instruct models. In addition to reproducing the paper’s LoRA-vs-DoRA setting, we also analyzed how DoRA interacts with instruction tuning by comparing base Llama against DoRA on base Llama, and Llama-Instruct against DoRA on Llama-Instruct. This allowed us to study both the qualitative improvement trend reported in the paper and the effect of the underlying base model on adaptation quality.

Methodology:

We re-implemented DoRA from scratch for a Llama-based causal language model. Our custom DoRALinear module freezes the pretrained weight and trains only lora_A, lora_B, and a learned magnitude vector. The low-rank update is

$$\Delta W = BA,$$

and the adapted weight is computed as

$$W' = m \cdot \frac{W + \Delta W}{\|W + \Delta W\|_c}.$$

Only adapter parameters are trained, and checkpoints store adapter weights rather than the full model.

Our implementation progressed in stages. First, we used **Llama-3.1-8B** with Alpaca-style prompts and response-only loss masking using -100. Following the paper’s LLaMA commonsense setup, we injected DoRA into

q_proj, k_proj, v_proj, up_proj, down_proj

We also tested attention-only adaptation to check whether modifying only attention projections was enough, or whether MLP projections were needed for stronger task adaptation.

Second, we varied rank, learning rate, batch size, epochs, and maximum sequence length. This helped us separate optimization/capacity issues from implementation or evaluation bugs. Third, we added **Llama-3.1-8B-Instruct** because early base-model outputs often failed to follow answer-format instructions; the instruct model helped test whether task alignment affected DoRA’s measured gains.

Finally, we revised evaluation. We first used greedy generation with exact matching, but this made scores highly sensitive to output formatting. Since the benchmarks are multiple-choice tasks, we switched to likelihood-based scoring, where each candidate answer is scored by average log probability. We also normalized dataset-specific labels such as

true/false, option1/option2, solution1/solution2, A/B/C/D/E.

Our experiments used the paper’s commonsense reasoning suite: BoolQ, PIQA, SIQA, HellaSwag, WinoGrande, ARC-e, ARC-c, and OpenBookQA.

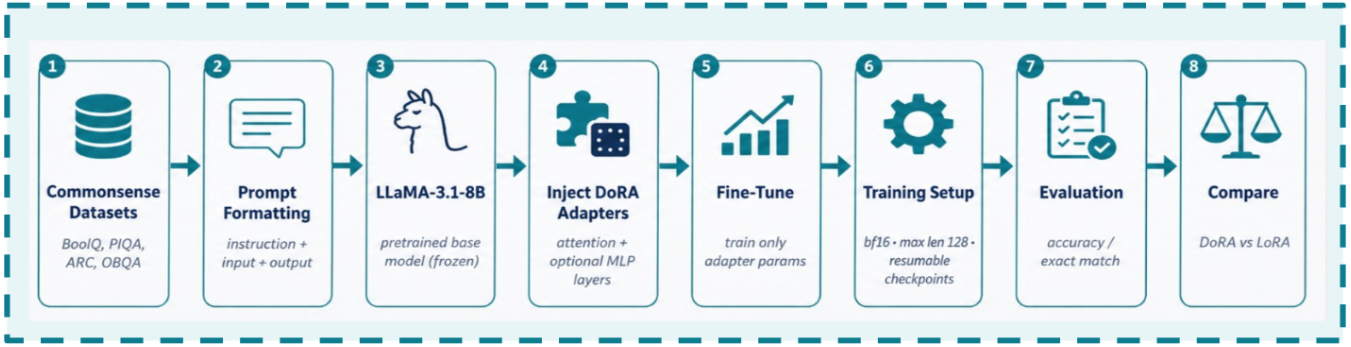


Figure 1: Step by step breakdown of training pipeline

Results & Analysis:

Our reproduction went through three main iterations. First, early runs produced near-zero accuracy across several datasets. We tried varying DoRA rank, learning rate, epochs, sequence length, and target layers to test whether the issue was insufficient adapter capacity or poor optimization. The problem persisted, and manual inspection showed that the model often generated verbose or off-format responses; in some cases, the correct answer appeared in the output but failed exact-match scoring.

Second, we traced the issue to the evaluation pipeline. The datasets used different label formats, greedy generation was brittle for multiple-choice tasks, and small prompt mismatches between training and inference added noise. To fix this, we normalized answer labels and replaced exact-match generation with likelihood-based multiple-choice scoring. After this change, the results became stable and interpretable.

Task	Llama-3.1-8B	Llama-3.1-8B (DoRA)	Llama-3.1-8B-Instruct	Llama-3.1-8B-Instruct (DoRA)
boolq	44.37	51.83	56.15	72.26
piqa	66.76	74.91	81.94	86.94
social_i_qa	42.22	56.42	68.63	78.15
hellaswag	28.73	45.78	69.00	83.84
winogrande	52.25	58.34	60.69	82.48
ARC-Easy	69.61	78.65	92.21	93.39
ARC-Challenge	53.50	63.82	78.84	83.19
openbookqa	45.20	57.13	75.60	84.00
Macro Avg	50.33	60.86	72.88	83.03

Table 1: Performance comparison across Llama-3.1-8B base, DoRA, instruct, and instruct DoRA variants.

Third, we compared base and instruction-tuned models with and without DoRA. Table 1 shows that DoRA improved every corresponding base model result. For Llama-3.1-8B, the macro average increased from 50.33 to 60.86, a gain of 10.53 points. For Llama-3.1-8B-Instruct, it increased from 72.88 to 83.03, a gain of 10.15 points.

The largest gains appeared on harder or more format-sensitive tasks. For example, base Llama improved from 28.73 to 45.78 on HellaSwag and from 42.22 to 56.42 on Social IQa. The instruction-tuned model also benefited strongly, especially on Winogrande, where accuracy increased from 60.69 to 82.48. Compared with the original paper’s LoRA results, our base-model DoRA scores were generally lower. However, Llama-3.1-8B-Instruct + DoRA was closer to or above the paper LoRA reference on several datasets, including BoolQ, PIQA, ARC-Easy, ARC-Challenge, and OpenBookQA. This suggests that absolute accuracy depends not only on the adapter method, but also on base model choice, instruction tuning, prompt formatting, and evaluation protocol.

Overall, our results reproduce the paper’s main qualitative finding: DoRA improves commonsense reasoning performance relative to the corresponding base model while remaining parameter efficient. At the same time, the reproduction showed that PEFT results are highly sensitive to implementation and evaluation details.

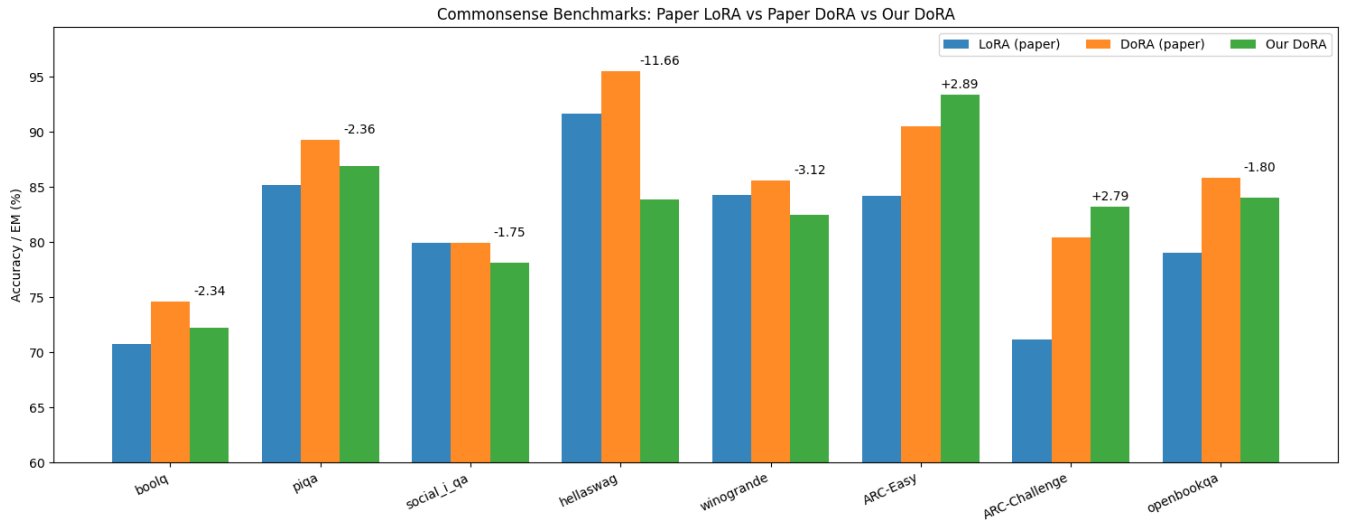


Figure 2: Comparison of LoRA (paper) vs DoRA (paper) vs Our DoRA across commonsense benchmarks.

Reflections:

This project showed us that reproducing PEFT methods requires both algorithmic understanding and careful experimental engineering. Although DoRA’s core idea is compact, reliable reproduction depended on adapter injection, checkpoint loading, prompt consistency, and evaluation design.

Our main progress came from iterating on the full pipeline rather than changing the core method: we debugged adapter implementation, fixed checkpoint loading, corrected prompt formatting, and moved from free-form generation to likelihood-based multiple-choice scoring.

A key lesson was that apparent model failure can come from evaluation mismatch rather than weak adaptation. We also found that base model choice matters: the instruction-tuned model followed answer-style prompts more reliably, while the base model made formatting and evaluation harder.

Given more time, we would complete a direct LoRA baseline, run larger-scale experiments across all eight datasets, compare against PEFT’s reference implementation, and perform ablations over rank, learning rate, sequence length, and target modules.

Overall, this project became an investigation of both DoRA and reproducibility in LLM fine-tuning.

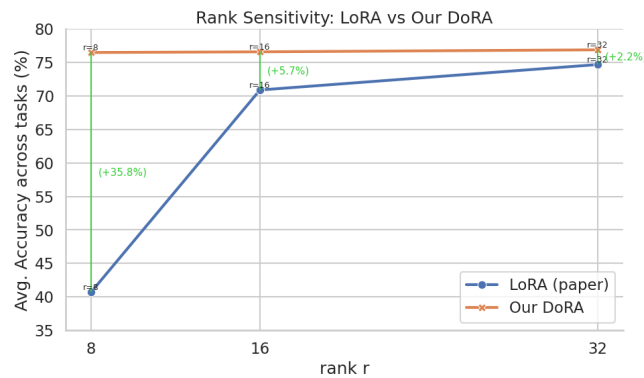


Figure 3: Average accuracy on all tasks (LoRA vs DoRA) across various ranks.

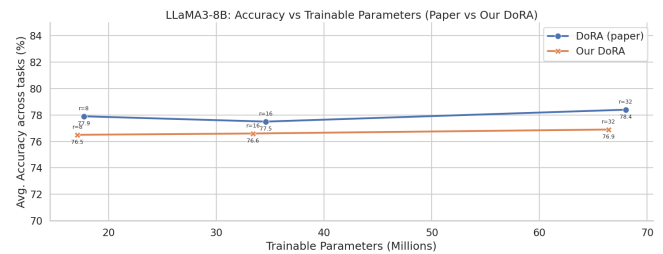


Figure 4: Consistency verification (DoRA paper vs our implementation) across various ranks.

References

1. Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen. *DoRA: Weight-Decomposed Low-Rank Adaptation*. arXiv:2402.09353, 2024.
2. Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. *LoRA: Low-Rank Adaptation of Large Language Models*. ICLR, 2022.
3. Meta AI. *Llama 3.1 Model Card*. 2024.
4. Thomas Wolf et al. *HuggingFace Transformers: State-of-the-Art Natural Language Processing*. EMNLP: System Demonstrations, 2020.
5. HuggingFace. *PEFT: Parameter-Efficient Fine-Tuning Library*. 2022–2024.
6. Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. *BoolQ: Exploring the Surprising Difficulty of Natural Yes/No Questions*. NAACL, 2019.
7. Yonatan Bisk et al. *PIQA: Reasoning about Physical Commonsense in Natural Language*. AAAI, 2020.
8. Maarten Sap et al. *Social IQa: Commonsense Reasoning about Social Interactions*. EMNLP, 2019.
9. Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. *HellaSwag: Can a Machine Really Finish Your Sentence?*. ACL, 2019.
10. Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. *WinoGrande: An Adversarial Winograd Schema Challenge at Scale*. AAAI, 2020.
11. Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. *Think you have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge*. arXiv:1803.05457, 2018.
12. Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. *Can a Suit of Armor Conduct Electricity? A New Dataset for Open Book Question Answering*. EMNLP, 2018.